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June 14, 2026

The Editor
Physica A: Statistical Mechanics and its Applications

Dear Editor,

I am pleased to submit our manuscript entitled “The Brody exponent as a measure of exclusion strength in 2D spatial point processes: evidence from manufactured surfaces, prime embeddings, and interferometric profilometry” for consideration for publication in Physica A.

Why Physica A? This paper develops a calibrated measurement framework for short-range exclusion in 2D spatial point processes—a statistical-physics methodology with applications spanning surface engineering, number theory, and optical metrology. The work bridges random matrix theory (Brody distribution) with established spatial point-process statistics, a topic well-suited to Physica A’s interdisciplinary statistical-mechanics readership.

Key contributions:

1. A necessary methodological correction: the 2D complete-spatial-randomness baseline for the Brody exponent is $\beta = 0.96 \pm 0.15$, not $\beta = 0$ as inherited from 1D applications.
2. A monotonic, reproducible β - $r_{\text{excl}}/\langle \text{NN} \rangle$ calibration established through synthetic hard-core and soft-core point processes, translating the dimensionless Brody parameter into a physically interpretable exclusion radius.
3. Rigorous density-thinning, embedding-robustness, spatial-permutation, and peak-detection sensitivity controls that isolate genuine exclusion from confounding effects.
4. Application to three mechanistically distinct systems: 58 manufactured surface topographies, 2D embeddings of prime numbers, and phase-extracted interferometric profilometry of a certified roundness standard.
5. A physical interpretation of the observed anti-clustering in prime embeddings: arithmetic divisibility constraints suppress short-range occupancy correlation, producing $\Delta D_2 > 0$ relative to density-matched random ensembles.

Rigour and reproducibility: All analyses are accompanied by bootstrap

confidence intervals, AIC-based model selection, Benjamini–Hochberg multiple-comparison correction, and power analysis. The complete 58-surface dataset, all analysis code, and a reproducible computational environment are archived in a public GitHub repository. Every numerical claim in the manuscript is verifiable against the deposited data.

Declarations: This manuscript is original, has not been published elsewhere, and is not under consideration by another journal. The author declares no competing interests. This is a single-author submission.

I confirm that the manuscript complies with Physica A’s formatting and ethical guidelines.

Respectfully,

Dawid Kucharski